

Mechanical, Electrical, and Ionic Behavior of Lithium-Ion Battery Electrodes via Discrete Element Method Simulations

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Herein, a discrete element method (DEM) approach is proposed to investigate the impact of the calendaring process on the electrical and ionic conductivities and on the adhesion strength of $\text{Li}[\text{Ni}_{1/3} \text{Mn}_{1/3} \text{Co}_{1/3}]\text{O}_2$ (NMC)-based electrodes. For this purpose, key correlations between the microstructure and these electrode-scale properties are established using the outcomes of the simulations and real experiments. In addition, the evolution of the structure and the development of mechanical stress are also studied numerically during electrochemical cycling, offering a closer insight into the intercalation mechanism. Finally, the impact of the initial noncalendered porosity on the electrode mechanical response is examined, showing that higher initial porosities lead to lower final porosities under same calendaring loads. Overall, this work demonstrates the potential of DEM simulations in improving the understanding of the microstructure and mechanics of lithium-ion electrodes.

1. Introduction

The widespread use of lithium-ion batteries (LIBs) has strongly encouraged scientific research on material, electrode, and cell level. To date, investigations have clearly confirmed the impact of the electrode microstructure on LIB performance metrics such as energy and power densities, cycling stability, cell life, and safety.^[1–4] However, the interactions between the electrode structural characteristics and the macroscopic response are still not fully understood mainly because correlations are interlaced and it is complex to analyze the parameters individually during a transient battery charge–discharge process. Taking this issue into consideration, numerical models and simulations have demonstrated to be a helpful tool with the additional advantage of saving raw material and time resources. In this context, Kespe and Nirschl^[5] investigated the influence of tailored particle

arrangement on the current density via a spatially resolved 3D microscale model. They demonstrated that a fraction of smaller active material particles located close to the separator leads to a decrease in average current densities within the electrolyte. Furthermore, an estimation of the diffusion resistance was presented by means of evaluating the electrode porosity and tortuosity. Also, focused on the role of the microstructure, Garcia et al.^[6] introduced a 2D spatially resolved model to describe the variations in electrochemical fields within LIBs. The model confirmed that a homogeneous narrow particle size distribution results in higher power densities, whereas aggregation phenomena shield active material particles and prevent them from a proper contact with the electrolyte matrix. Chen

et al.^[7] used a predictive model to investigate the course of electronic and ionic conductivities in $\text{Li}[\text{Ni}_{1/3} \text{Mn}_{1/3} \text{Co}_{1/3}]\text{O}_2$ (NMC) electrodes with the aim of uncovering the trade-off required for a maximum specific energy. Another aspect that has been partially considered in modeling the microscale is the volumetric expansion and contraction of active material particles during electrochemical cycling. This aspect is extremely critical because it implies a significant stress generation within the electrode that can damage the particle network and consequently affect the electrochemical performance of the LIB.^[8–12] Within this framework, 2D particle-resolved models have been developed,^[6,13] and furthermore, 3D battery models have been used to evaluate the empirical Bruggeman exponent and investigate the effect of the electrode microstructure on the battery performance.^[14] Trembacki et al.^[15] used tomography data of an NMC-based cathode to assemble a mesh and extract electrode-scale effective properties via finite element continuum-scale simulations considering the lithiation-dependent mechanical swelling of the particles. Although the aforementioned studies have provided general guidelines regarding the impact of the microstructure, the great majority were solved either by finite element methods or finite volume methods. Furthermore, the mechanical behavior of active material particles was not completely considered and, in some cases, the presence of the binder-additive matrix was disregarded. In general lines, the coating thickness, porosity, and the total mass loading have been defined as the main descriptors to characterize a composite electrode.^[16] Nevertheless, the interdependencies are much more complex and additional parameters or even directional values must be unraveled to fully understand the role of the microscale

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during the cell operation. In a previous study, we proposed a discrete element simulation (DEM) method to analyze the calendaring process directly from the mechanical behavior of the additive-binder matrix and single active material particles to the macroscopic mechanical response of NMC-based electrodes.^[17] The validated simulations brought the possibility to characterize additional structural and mechanical features of the electrodes that are difficult or not possible to obtain via experiments. Taking this evaluation procedure into consideration, the present study aimed to complete this line of work as shown in **Figure 1**. The outcomes of the DEM simulations were considered to establish a direct relationship between the microscale and the overall response of the electrode after the calendaring process. In particular, the focus was set on modeling the electrical and ionic conductivities as well as the adhesion strength, which are the key properties that have already been proved to have a direct impact on the performance and life of a battery.^[18–20] In order to do so, the outcomes of our first publication were considered together with experimental measurements to validate the derived correlations. Once the modeled properties could be validated, the simulated cathodes were used to investigate the effect of lithium intercalation and deintercalation on the evolution of the microstructure and the global stress development. Finally, an additional simulation study was conducted on the influence of the initial noncalendered porosity on the mechanical response of the electrode during the calendaring process. The modeled electrical and ionic conductivities along with the adhesion strength were also evaluated, and their correlation to this structural parameter could be elucidated.

2. Results and Discussion

2.1. Predicting Electrode-Scale Properties from the Microscale

A well-known method to accurately link the microscopic and the macroscopic scales is the homogenization theorem,

which is concisely presented in previous studies.^[21,22] The homogenization theory aims to define effective material properties at the representative volume element (RVE) scale by using constitutive relationships that relate variables at the global or macroscopic level. In the present study, the constitutive relationships on the global or electrode scale comprehend the electrical conductivity, ionic conductivity, and adhesion strength.

2.1.1. Electrical Conductivity

The electrical conductivity plays a fundamental role in the performance of a battery cell as it accounts for the conduction of the electrons within the porous electrode.^[23,24] Because the electrode is not a homogenous isotropic composite, the electrical conductivity cannot be reduced to a single scalar value. In fact, during the charging and discharging process of the battery, electrons are bound to move from the current collector in the direction toward the separator region and vice versa; hence, for modeling, it is important to consider the normal component of the electrical conductivity. In the field of granular matter, Batchelor and O'Brien already proposed an expression to describe the connectivity of a single grain with the rest and how it can establish the thermal conduction of the equivalent continuum.^[25] Based on this work, the following constitutive relationship was formulated to model the electrical conductivity within the RVE in normal direction, perpendicular to the current collector

$$k_{el} = C_{el}(1 - \varepsilon) \frac{CN}{\pi} 3F_{zz} \frac{r_c}{r_p} k_p \quad (1)$$

The first term in the equation, C_{el} accounts for a proportionality factor that could be calculated taking into consideration the experimental data. The solid fraction in the simulated RVE, $1 - \varepsilon$, is considered along with the coordination number CN, i.e., the number of contacts per active material particle. F_{zz} corresponds

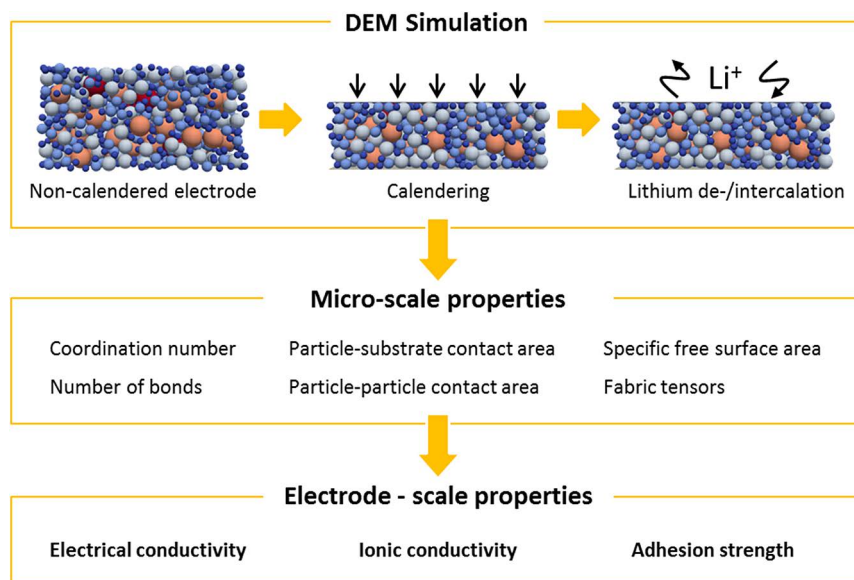


Figure 1. Schematic of the overall concept of this study. DEM simulations were used to simulate the calendaring and electrochemical cycling processes. Microscale properties were derived from the simulations to define electrode-scale properties.

to the diagonal fabric tensor in the normal direction (perpendicular to the current collector). This parameter accounts for the distribution of particle contacts, and it has to be taken into consideration here because the electrical conductivity is a directional-dependent quantity. That is, it is not only important that the particles are well connected but also that the directionality of these contacts are favorably in the normal direction, perpendicular to the current collector. The ratio r_c/r_p represents the relationship between the average contact radius and the average particle radius. All these parameters are defined and calculated as fully explained in our first publication.^[17] k_p stands for the conductivity of a single NMC particle, which is taken from the work of Amin and Chiang^[24] and equals to $10^{-5} \text{ S cm}^{-1}$.

2.1.2. Ionic Conductivity

In addition to the electrical conduction, ionic conduction through the electrode is necessary to complete the electrochemical reaction. The ionic conductivity in a macroscopic porous electrode can be analytically calculated based on the porosity (ϵ) and tortuosity (τ) as^[19]

$$k_{\text{ion,analytical}} = k_{\text{ion}}^{\text{bulk}} \frac{\epsilon}{\tau} \quad (2)$$

where $k_{\text{ion}}^{\text{bulk}}$ corresponds to the bulk ionic conductivity of the electrolyte. The tortuosity τ is an essential electrode property, especially for high power density applications, as it represents the feasibility of lithium ions to move along the porous electrode. It can be determined by the well-known Bruggeman relation^[26]

$$\tau = \epsilon^{-0.5} \quad (3)$$

Taking both equations into account, the ionic conductivity in the macroscale is established to be proportional to $\epsilon^{1.5}$. Although this relationship has been widely used in many prior porous battery models, it is clear that the tortuosity of a given material cannot be uniquely determined by porosity.^[26–28] In fact, the Bruggeman relation is valid for continuous phases mixed with a second phase of uniform-sized insulating spheres. This implies that the ionic conductivity is not assured to be 100% accurately predicted for the cases presented in this study.

Considering the pure definition of the ionic conductivity, it was determined that the diffusivity of lithium ions had to be directly proportional to the inner free surface area and the porosity of the simulated electrode. Because the ionic charge carriers need to traverse in-through direction, the directionality of the particle contacts is also relevant. Particle contacts in the direction parallel to the current collector hinder the pathways for ion transport, whereas contacts arranged in-through direction are favorable. Taking these aspects into consideration, we defined the constitutive relationship for the ionic conductivity as

$$k_{\text{ion}} = C_{\text{ion}} FSA \epsilon \frac{CN_{zz}}{CN_{xx}} k_{\text{ion}}^{\text{bulk}} \quad (4)$$

where C_{ion} is the proportionality factor; FSA is the specific free surface area, i.e., the total surface area of all particles that is not involved in any contact; and CN_{zz} and CN_{xx} are the coordination numbers in the normal direction (perpendicular to the current

collector) and the direction parallel to the current collector, respectively. These parameters are obtained from the simulations as fully explained in the study by Sangrós Giménez et al.^[17]

2.1.3. Coating Adhesion Strength

The coating adhesion strength is a very important mechanical property as it affects the integrity of the electrode that influences the proper handling during cell production and the electrochemical lifetime of the battery cell.^[20] The adhesion strength accounts not only for the proper contact between the coating and the current collector but also for the cohesion within the coating, i.e., the performance of the binder is therefore fundamental, as already reported in the study by Jaiser et al.^[29] In this regard, the constitutive relationship for the coating adhesion strength was calculated in this study by considering a proportionality factor, C_{ad} ; the area of the current collector in contact with NMC particles ($A_{\text{CC-NMC}}$) related to the total area of the current collector (A_{CC}); the average bond force (F_b) that represents the particle-particle cohesion force brought by the additive-binder matrix in the simulation; and the average number of bonds per particle (B_p) within the RVE

$$\sigma_n = C_{\text{ad}} \frac{A_{\text{CC-NMC}}}{A_{\text{CC}}} F_b B_p \quad (5)$$

2.2. Impact of Calendering Degree on Electrode Properties and Electrochemical Cycling

In a previous study, the mechanical behavior of Li-ion electrodes during the calendering process was investigated via DEM simulations, which were calibrated and validated with the support of real calendering experiments. Starting from a noncalendered electrode with a porosity of 0.522, four calendering degrees were accomplished experimentally. To simulate the behavior of the electrode coating, an elasto-plastic contact model based on the work of Thornton and Ning^[30] was combined with a bond model.^[31] The contact model captured the mechanical behavior of NMC particles, while the bond model represented the additive-binder matrix holding the particles together. An overall explanation about the simulation procedure can be found in the Computational Section, while further precise details can be found in the study by Sangrós Giménez et al.^[17] **Table 1** summarizes the most important structural and process-related aspects

Table 1. Relevant structural and process parameters of the simulated calendered cathodes and the initial noncalendered cathode.

Electrode	Porosity [-]	Coating density [g cm ⁻³]	Calendering stress [MPa]
Noncalendered	0.522	2.13	—
C1	0.417	2.61	19.82
C2	0.368	2.80	42.71
C3	0.305	3.15	79.13
C4	0.270	3.49	159.37

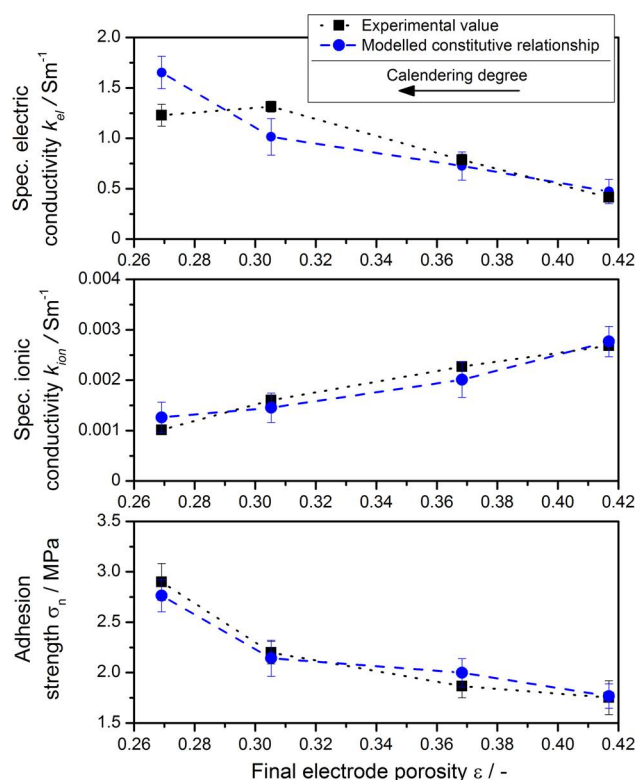


Figure 2. Specific electrical conductivity, specific ionic conductivity, and adhesion strength as a function of the final electrode porosity after calendering. Experimental and simulated results are presented.

for the calendered cathodes obtained via the DEM-based simulation strategy.

The specific electrical conductivity and the adhesion force of the calendered cathodes were measured experimentally, whereas the specific ionic conductivity was derived from the Bruggeman relation considering the final experimental electrode porosity. Using these measured values and the outcome of the simulations, the proposed constitutive relationships were validated. **Figure 2** shows the experimental and the calculated values for the four calendered electrodes using Equation (1–4).

The specific electrical conductivity was fitted with a proportionality factor C_{el} of 0.665. Overall, it can be seen that calendering promotes an increase in the specific electrical conductivity if the final electrode porosity is higher than $\approx 30\%$. Within this porosity range, the qualitative trend can be described by the proposed constitutive relationship. Therefore, the specific electrical conductivity is directly dependent on the number of particle contacts in the direction perpendicular to the electric current and the ratio between the contact and the particle radius. If the final electrode porosity achieved after the calendering process is lower than 30%, a decrease in the specific electrical conductivity can be discerned experimentally. One reason for this reduction could be the increase in the particle breakage. To assess this aspect, a host of NMC particles with different diameters were experimentally analyzed via nanoindentation. A stress-based criterion was adopted as NMC is a stiff material and the acting stress is not reduced by plastic deformation prior to the breakage event (brittle

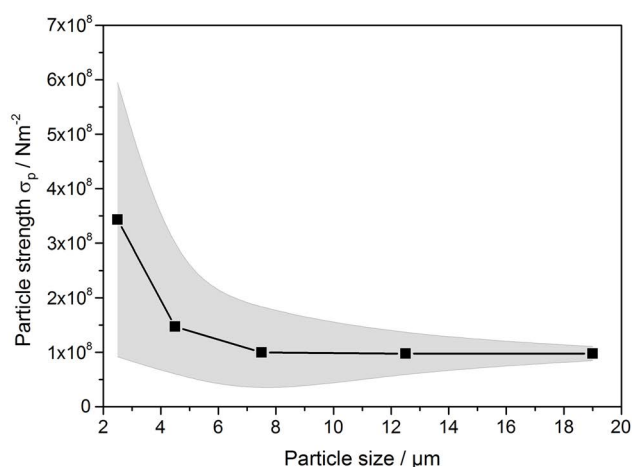


Figure 3. Average particle strength as a function of the size for NMC particles measured via nanoindentation.

fracture). This criterion implies that an interface debonding occurs when the local normal stress and the interface reaches the interfacial strength. Thus, the first step was to calculate the particle strength. This was performed by compressing single secondary particles till breakage and determining the stress on the indenter at this point (see Experimental Section). As shown in **Figure 3**, the particle strength decreases with the particle size. Hence, the stress acting on each individual particle was calculated in the simulations and compared with the experimental particle strength taking into consideration the size of the analyzed particle. Accordingly, if the numerical stress was greater than the experimental particle strength, the NMC particle was determined to be fractured in the DEM simulation. The resulting portion of broken particles within the structure was evaluated for every electrode at the position under the maximum stress (end of compaction, immediately before the pressure starts to be released).

Table 2 shows that an insignificant portion of particles broke during calendering even at high levels of stress (1.52% broken particles within electrode C4). Consequently, the decrease in electrical conductivity is not mainly a result of particle fracture as particles were able to withstand the stress levels induced during calendering without breaking.

In fact, the reason for the reduction in electrical conductivity may lie primarily in the loss of mechanical integrity within the electrode. It has already been reported that high levels of calendering can cause fundamental changes in the conductive network. In particular, it can induce contact losses in the electric pathways, leading to an increase in the electrical resistance.^[4] Taking this into consideration, our next step will be to further refine the modeling of this parameter by considering the conductivity of the additive–binder matrix as well.

Table 2. Fraction of broken particles for the calendered electrodes simulated via DEM.

Electrode	C1	C2	C3	C4
Broken particles [%]	0	0.09	0.54	1.52

The specific ionic conductivity was calculated in the simulations considering that the bulk ionic conductivity of the electrolyte was the same for all cases and equal to 0.01 S cm^{-1} . The proportionality factor C_{ion} equaled 0.534. In general lines, the decrease in ionic conductivity derived from the Bruggeman relation with increased calendering levels can be qualitatively well reproduced by the proposed constitutive relationship. Thus, it can be stated that the decrease in the free surface area impacts the area that is electrochemically active for Li-ion insertion and removal. Furthermore, the directionality of particle contacts also affects the ionic conductivity of the electrode. Likewise, the suggested relationship of structural and mechanical parameters to describe the adhesion strength can be validated with the experimental data and a proportionality factor of C_{ad} of 0.755. The trend can be qualitatively achieved, which demonstrates that the mechanical integrity of the electrodes depends on the interface between the particulate assembly and the current collector as well as on the performance of the binder, which is represented through bonds between particles in the simulation.

Taking into consideration the changes in volume and mechanical properties of NMC particles, the evolution of the electrode structure was explored during lithium-ion deintercalation and intercalation in subsequent simulations. This is of great interest due to the consequent impact on the transport and mechanical properties, which at the same time has an effect on the performance and life cycle of the battery cell.^[11,32] To this end and directly after the calendering simulations of these electrodes, the resulting structures were subjected to electrochemical cycling as fully explained in the Computational Section. The corresponding structural characteristics were determined along with the course of mechanical stress. Specifically, **Figure 4a** shows the evolution of the coordination number during the first five cycles for all the electrodes, i.e., the average number of contacts per

particle. Because the swelling and contraction of NMC particles was accompanied by changes in Young's modulus corresponding to the state of charge,^[32] the curves do not present a homogeneous form. In all cases, the maximum values of the coordination number were obtained at the beginning immediately after the calendering simulation was concluded. During the first half cycle, lithium concentration within NMC particles decreased, leading to a contraction of the particles and hence, a reduction of the coordination number for all electrodes. Once the lithium concentration started increasing (second half of the cycle), the particles swelled and, thus, the coordination number increased. This trend is repeated throughout the five electrochemical cycles. It is also discerned that the coordination number achieved at the end of every cycle tends to decrease. This is of great relevance because it confirms that one degradation mechanism of electrode cycling is in fact the structural disintegration and the loss of particle contacts within the electrode coating, which can be studied via DEM simulations. Furthermore, the additive-binder matrix plays an important role in the mechanical integrity of the electrode because it holds the particles together and provides connection to the current collector during charging and discharging.^[33] As mentioned earlier, this is represented through bonds between particles in the simulation. Because the bond model is aimed to represent the mechanical performance of the additive-binder matrix, this parameter can be regarded as a scale to assess the connectivity within the structures. The bond model, fully explained in the study by Sangrós Giménez et al.,^[31] transfers normal and tangential forces to the particles. Furthermore, if a critical strength, also called the ultimate strength, is reached, the bond gets broken and the corresponding forces and momentum are eliminated from the simulation. As published in the study by Sangrós Giménez et al.,^[17] the area-related bond stiffness to simulate the calendered

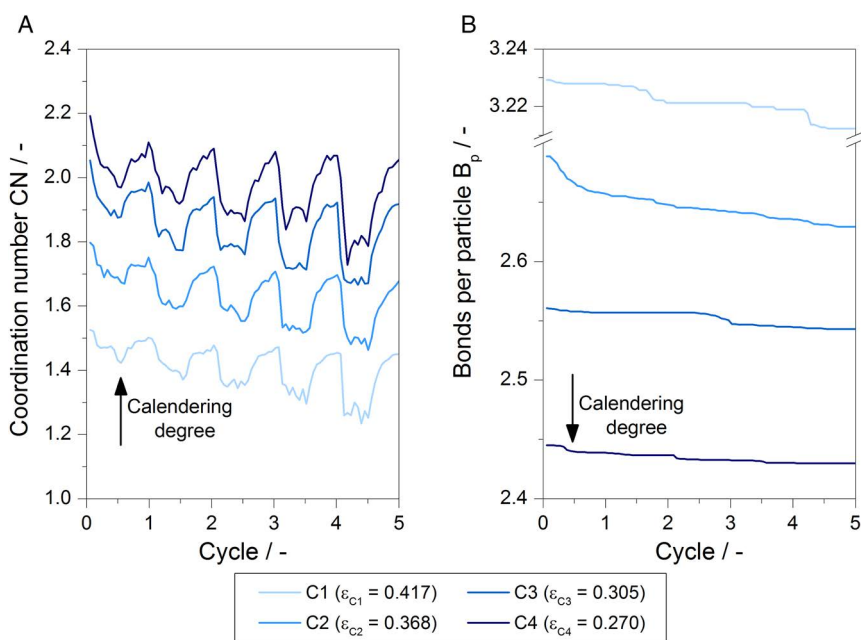


Figure 4. a) Coordination number along the charging–discharging process for the four calendered electrodes. b) Number of bonds per particle along the charging–discharging process for the four calendered electrodes.

cathodes was $6 \times 10^{12} \text{ N m}^{-3}$, and the critical stress to define the bond breakage was $2 \times 10^{13} \text{ N m}^{-2}$. Bearing this in mind, Figure 4b shows the number of bonds per particle along the five cycles for all the four simulated electrodes. It is discernible that directly after the calendaring simulation, electrode C1 with a porosity of 0.417 presented the higher number of bonds per particle, around 3.23. The greater the calendaring line load, the higher the number of broken bonds at the end of the process. Consequently, the number of bonds per particle at the beginning of the electrochemical cycling for electrode C4, with a porosity of 0.270, was comparatively lower, around 2.45. Due to the levels of mechanical stress already arisen at the calendaring process, electrodes C3 and C4 do not exhibit a significant loss in the number of bonds per particle during the five cycles. However, it can be perceived that this parameter decreases substantially for electrodes C1 and C2 because the ultimate yield strength was not reached along the calendaring process but during the charging–discharging process for a relatively high amount of bonds. Overall, it can be stated that the DEM simulation procedure corroborates and helps to quantify the loss of connectivity within the particulate electrode structure.

It has already been reported that during the performance of a LIB, stress develops as a consequence of Li insertion and removal.^[6,34,35] Large intercalation levels of stress can be critical because they can lead to capacity fade and eventual battery failure. Mukhopadhyay and Sheldon already presented an in-detail description of the stress development within electrode materials.^[8] In their work, the major reasons for the evolution of stress during electrochemical cycling were grouped into several mechanisms. Within the framework of the present study, we considered mainly the constraining effects of neighboring active material particles and particles in contact with the current collector. For this purpose, the viral stresses on individual particles were calculated first, as explained in the Computational Section. Second, the total macroscopic stress was averaged over the total volume of the electrode and correlated to the Li concentration gradient. Figure 5 shows the evolution of the mechanical stress within the electrodes as a function of the state of charge

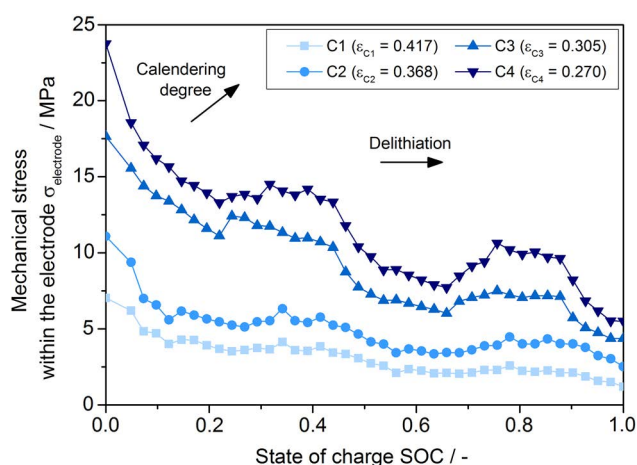


Figure 5. Stress evolution along the first electrochemical cycle for the previously calendared electrodes.

during the first delithiation, where 0% represents the pristine NMC state and 100% corresponds to $\text{Li}_{1/2}[\text{Ni}_{1/3} \text{ Mn}_{1/3} \text{ Co}_{1/3}]\text{O}_2$.

It can be discerned that electrode C4 presents the highest mechanical stress in comparison with the other structures. Due to its low porosity after calendaring ($\epsilon_{C4} = 0.270$), the number of particle–particle contacts within this structure is higher, and thus, the particle constraints and the corresponding developed stresses are higher. In general, compressive mechanical stresses decrease along the delithiation process as a result of the reduction in particle volume. The curves obtained in the present study can be compared with the experimental ones reported in the study by Nadimpalli et al.,^[36] measured by monitoring the substrate curvature with a multibeam optical sensor setup. In this experimental research, different regimes of stress evolution within the cathode could also be distinguished and related to volume changes associated with phase transformations. In addition, the binder can help to buffer stresses in the porous composite electrode. In fact, the standard polyvinylidene fluoride (PVDF)-based binders can accommodate stresses of $\approx 10\%$.^[37,38] To investigate the effect of the binder mechanical properties on the evolution of the stress within the electrode, the same simulations were performed again but without applying the bond model. Therefore, for these simulations, only particle–particle interactions were responsible for the stress generation. Exemplarily, Figure 6 shows the stress evolution within the electrodes C2 ($\epsilon_{C2} = 0.368$) and C4 ($\epsilon_{C4} = 0.270$), considering the bond model and without its consideration. It can be seen that the bonds can capture the reported behavior as the achieved levels of stress without the bond model are moderately higher. Hence, the simulation method presented here is believed to be valid to investigate structural parameters and stress metrics with the aim of pointing to better electrode structures.

2.3. Impact of the Initial Noncalendered Porosity on the Electrode Behavior during Calendaring

The next step was to numerically study the impact of initial porosity, i.e., the porosity of the noncalendered structure, on

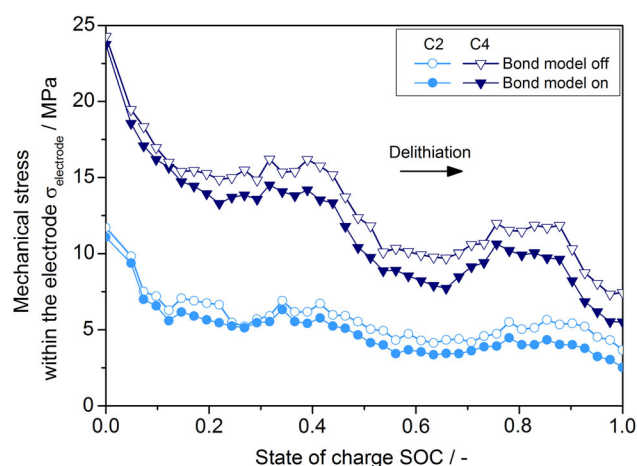


Figure 6. Stress evolution along the first electrochemical cycle for the previously calendared electrodes C2 and C4, considering the effect of the binder (solid lines) and without the effects of the binder (dashed lines).

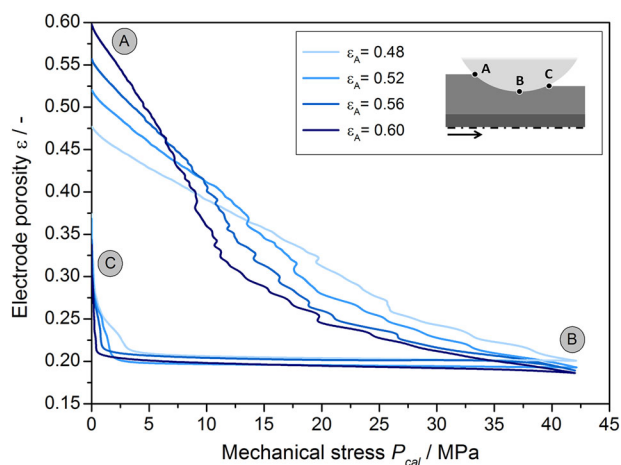


Figure 7. Evolution of electrode porosity as a function of the applied mechanical stress for the simulated electrodes with different initial non-calendering porosities.

the structural, mechanical, and electrical response of the electrode during calendering. The porosity of the initial structures was varied from 0.48 to 0.60 by changing the number of particles in the simulation RVE while assuring the same particle size distribution and electrode thickness. Likewise, the formulation as well as the additive-binder matrix and particle properties were held constant for all cases. The aimed calendering mechanical stress was selected at 42.71 MPa. **Figure 7** shows the evolution of the electrode porosity along the calendering process for all simulated structures. Starting from the noncalendered electrode (position A in the figure), the porosity decreases due to the compaction action of the plate. Once the aimed mechanical stress is achieved (position B), the plate begins to move upward. The final porosity (position C) is moderately higher than the porosity in position B as an effect of the elastic recovery of the particles and the binder. The higher the initial porosity, the more the free space around the NMC particles. Thus, the constraint they have to overcome is lower as they can be easily rearranged during compression. As a result, the porosity decreases more rapidly and reaches lower values at the end of the compaction stage (position B). The final porosity achieved after the unloading of the plate (position C) is lower for higher values of noncalendered porosities.

Figure 8 shows more clearly the porosity values in positions B and C as a function of the initial electrode porosity. Consequently, to obtain highly compacted NMC-based electrodes via calendering without having to apply extremely high line loads, it is advisable to target high porosities already at the coating and drying process.

Analogously to the previous section, the specific electrical conductivity, the specific ionic conductivity, and the adhesion force were calculated via the validated correlations (Equation (1–4)) and the fitted proportionality factors. In order to do so, the structural and mechanical parameters were determined for all electrodes after the calendering simulation was performed. **Figure 9** shows the corresponding values for the four simulated electrodes. It can be seen that the initial noncalendered porosity has an impact on the mechanical and transport properties of the electrode as a

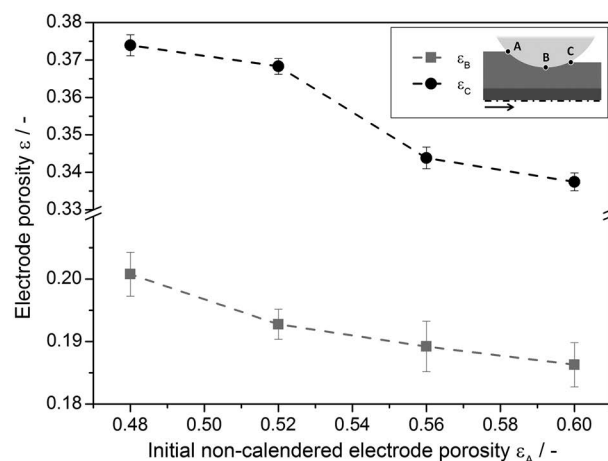


Figure 8. Values of porosity at the position of the maximum mechanical stress (position B) and at the end of the calendering process (position C) as a function of the initial noncalendered porosity (position A).

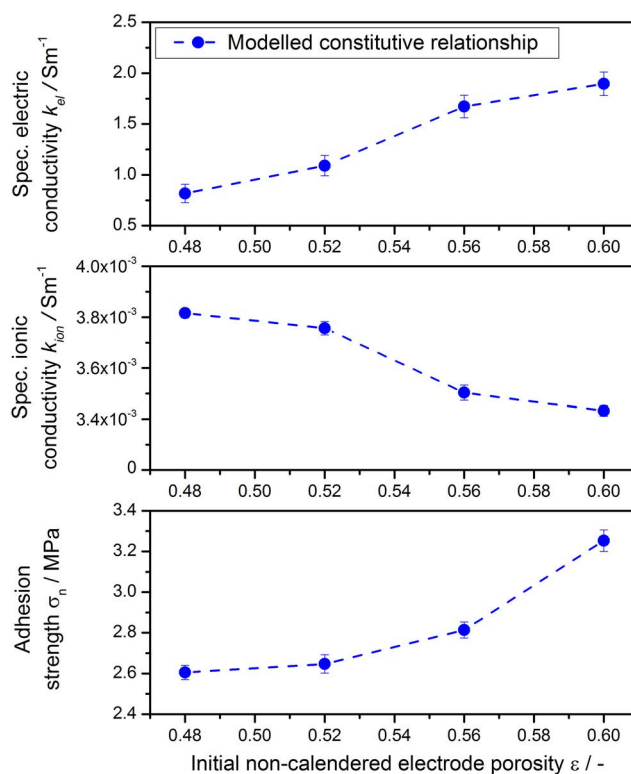


Figure 9. Modeled specific electrical conductivity, specific ionic conductivity, and adhesion strength as a function of the initial noncalendered electrode porosity.

result of the connectivity within the structure. Higher values of initial noncalendered porosity lead to lower values of porosity after calendering, which translates into a higher coordination number and an increased contact area among the particles and between the particles and the current collector. Consequently, the specific electrical conductivity increases,

whereas the specific ionic conductivity decreases due to a reduction in the free surface area. Another important aspect is the increase in the bond forces that lead to greater adhesion strength for higher initial noncalendered porosities.

3. Conclusions

This work, which constitutes a second part of what is aimed to be a comprehensive numerical study on the calendering process, offers the possibility to give an insight into the microscale of LIB electrodes to understand their structural and mechanical response. In particular, the evolution of the electrode structure was analyzed during the calendering step as well as along the lithium intercalation and deintercalation process via DEM simulations. The aim was to use the outcome of the simulations to model key electrode properties, namely the electrical and ionic conductivities and the adhesion strength. With the support of experiments performed on real NMC-based electrodes, the constitutive relationships to model these electrode properties could be validated. In particular, the specific electrical conductivity was proved to be dependent on the number of particle contacts in the direction perpendicular to the electric current and the ratio between the contact and the particle radius. For the highest level of calendering degree, the numerical value diverged from the experimental outcome, which could be explained with the loss of mechanical integrity within the electrode structure as the particles were demonstrated not to break during the calendering process. Furthermore, it was shown that the ionic conductivity is dependent on the free surface area and the directionality of particle contacts, whereas the adhesion strength depends on the interface between the particulate assembly and the current collector as well as on the performance of the binder, which was represented through bonds between particles in the simulation. In subsequent simulations, the effects of electrochemical cycling were studied by taking into account the changes in volume and mechanical properties of NMC particles. It could be demonstrated that DEM simulations can help to gain additional information regarding the mechanical integrity of the electrode and the development of mechanical stress within the structure. The first was examined by tracking the coordination number, that is, the number of contacts per active material particle, and the number of bonds per particle, which decreased along the five simulated cycles. The evolution of the mechanical stress as a function of the state of charge was presented, and the ability of the binder to buffer stress could be confirmed to be captured in the simulations via the bond model. In addition, a simulation study was conducted on the effect of the initial noncalendered porosity on the mechanical response of the electrode during the calendering process. It could be unravelled that to achieve highly compacted NMC-based electrodes, it is advisable to target high porosities after the coating and drying processes. The final electrode structure depends on the initial porosity, which has an impact on the electrode mechanical and transport properties after applying a constant calendering normal stress. In fact, with high values of initial noncalendered porosity, the specific electrical conductivity increases due to an increment in the coordination number, whereas the specific ionic conductivity decreases due to a reduction in the free surface area. The increase in

the bond forces leads to greater adhesion strength for higher initial noncalendered porosities. Overall, this numerical study gives an insight into the impact of the electrode microstructure and micromechanics, which can help to improve electrode design. Our next step is to further analyze initial electrode properties and to refine and include further constitutive relationships, which will lead to comprehend the calendering and the intercalation processes to a large extent.

4. Experimental Section

For the validation of the calendering simulations and the definition of the correlations, an NMC-based cathode was manufactured and subjected to different calendering line loads. Conductive graphite (2 wt%) and carbon black (4 wt%) were used as conducting additives and PVDF (4 wt%) as a binder. The comprehensive description of the production process can be found in our first publication, whose focus was set on the simulation of the calendering process itself.^[17]

Specific Electrical Conductivity: The specific electrical conductivity was measured using the method and equipment described in the study by Westphal et al.^[39] The applied current was 10 mA, the contact stress was 350 kPa to ensure that the electrode was in contact with the stamps, and the measuring time was set at 3 s.

Coating Adhesion: To measure the coating adhesion, the method and equipment were used following the work by Haselrieder et al.^[20] The start phase was set at 1.14 s, and the compression rate was established at 0.75 mm min⁻¹. Once the stress was achieved, it was kept constant for 30 s. After the dwell time, the pull-off phase occurred at a rate of 100 mm min⁻¹ and the maximum tensile force was detected.

Particle Strength: Nanoindentation experiments on single NMC particles were performed to determine their particle strength (TribolIndenter TI 900 from Hysitron, Inc Minneapolis) according to the work by McDowell and Bolton.^[40] A steel flat punch nanoindenter with a diameter of 50 µm was used, and the compression rate was set at 30 nm s⁻¹. A host of 80 particles were stressed until fracture occurred. At this point, the force was recorded and the particle strength σ_p was calculated individually for every particle

$$\sigma_p = \frac{F_{\text{break}}}{\pi r_p^2} \quad (6)$$

where F_{break} corresponds to the measured force required to reach particle fracture and r_p is the radius of the particle.

5. Computational Section

The first contribution of our work dealt with the characterization of NMC particles with the aim of representing their mechanical behavior correctly.^[17] The experimental nanoindentation results claimed the requirement for using an elasto-plastic contact model, which clearly exerted a great impact on the macroscale. DEM simulations of the calendering process were performed, and the model was calibrated in combination with a bond model to represent the binder-additive matrix.^[31] The calibration and validation of the simulation procedure were possible due to the performance of real calendering experiments. For further information about the simulation procedure and obtaining of the parameters, readers are referred to the study by Sangrós Giménez et al.^[17] In general lines, **Figure 10** shows the simulation schematically. An RVE of 150 µm × 150 µm × 78.15 µm (78.15 µm is the thickness of the noncalendered electrode) was chosen with periodic boundaries in x and y directions (position A). On the bottom, a plate with aluminum properties was placed to represent the current collector. On the top, a plate representing the calendering roll compressed the structure till the maximum normal stress was achieved (position B). Once this was accomplished, the pressure was released and the final electrode structure was obtained (position C).

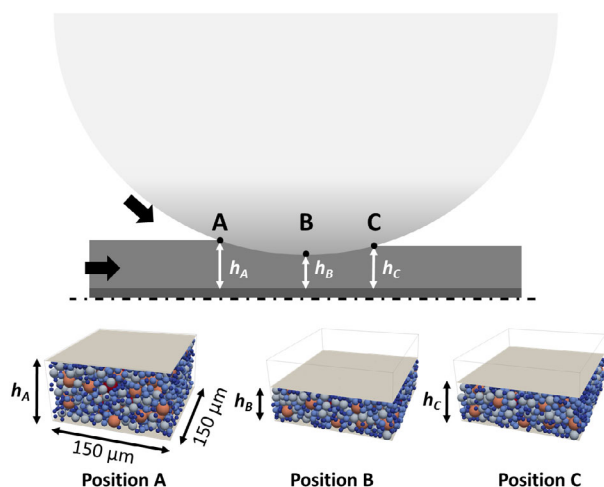


Figure 10. Schematic of the DEM simulation regarding the calendaring process. Starting from position A, the electrode is compressed till position B is achieved. Finally, due to the elastic recovery, the final structure is obtained (position C).

In total, 2237 particles were simulated. The particle size distribution of the real produced NMC electrode was measured and brought into the simulation to assure a proper representation of the structure. During the calendaring simulation, changes in structural and mechanical features took place on the microscale. In particular, we defined and studied the following parameters: the specific free surface area, the coordination number of NMC particles, the number of broken bonds that represent the deterioration of the binder, the contact area between NMC particles and the aluminum current collector, the fabric tensors, and the stress evolution within the electrode. In this second contribution, this knowledge is used to evaluate the constitutive relationships concerning the electrical and ionic conductivities and the adhesion strength. Furthermore, the focus of the present work was set on the structural and mechanical aspects arisen not only during the calendaring process but also along the subsequent Li-ion deintercalation and intercalation. This additional part was simulated by taking into account the volumetric change of NMC particles and the evolution of their mechanical properties induced by the lithium concentration gradient. Although the volume increase of NMC particles was reported to be only around 1.54% when lithiated from $x = 0.5$ to $x = 1$, x being the stoichiometric value of lithium in NMC, it can trigger different phenomena.^[15,41,42] Recent investigations have shown that the stress induced upon lithiation may lead to particle fracture,^[43] structural disintegration, and even mechanical failure, which significantly decrease the electronic and ionic conduction, reduces cycle efficiency, and ultimately result in battery failure.^[8,10,44] Bearing this in mind, we aimed to simulate this process via DEM and to decouple the effects of structural and binder-related mechanical parameters to obtain a comprehensive understanding. The micromechanics of lithium deintercalation and intercalation were simulated by considering the following two aspects: the volume swelling and compression of NMC particles and the corresponding variations in elastic modulus due to phase changes in the oxide during the process. The radii of NMC particles were adjusted according to the experimental study by Kam et al.^[45] In this work, the crystal lattice unit cell volume was measured along the lithiation of NMC333 with ex situ synchrotron X-ray diffraction measurements. As a result, a linear correlation between particle strain and Li-ion concentration could be identified. Because of the small directional dependency of volume change, we considered the lithiation-induced volume correction to be isotropic within the electrode. Thus, the lithiation-dependent strain in each coordinate direction is equivalent. Likewise, the Young's modulus of NMC particles was adapted along the lithiation state as shown in the study by Xu et al.^[32] The progressive reduction of the

elastic modulus during delithiation was implemented starting with a value of 142.5 GPa for pristine NMC and ending with a value of 111.6 GPa for $\text{Li}_{0.5}[\text{Ni}_{1/3} \text{Mn}_{1/3} \text{Co}_{1/3}]\text{O}_2$. The swelling of particles was constrained by the neighboring particles that were expanding as well. The mechanical stress evolution was investigated during the process and calculated in terms of microscopic forces. Overall, the procedure consists on determining the stress tensor of each individual particle as the sum of the stress due to direct particle contacts and bond interactions

$$\sigma_{ij} = - \left[\frac{1}{2} \sum_{n=1}^{N_p} (r_{1i} F_{1j} + r_{2i} F_{2j}) + \frac{1}{2} \sum_{n=1}^{N_b} (r_{1i} F_{b,1j} + r_{2i} F_{b,2j}) \right] \quad (7)$$

The first term corresponds to the pairwise energy contribution, where n loops over the N_p neighbors of particle 1, r_{1i} and r_{2i} are the positions of the two particles in the pairwise interaction, and F_{1i} and F_{2i} are the forces acting on the corresponding particles resulting from the particle interaction. The second term corresponds to the bond contribution, where n loops over the N_b bonds of particle 1 and $F_{b,1i}$ and $F_{b,2i}$ are the forces associated with the bond. Once the stress tensor is computed for every particle, the principal stresses, i.e., σ_{xx} , σ_{yy} , and σ_{zz} , are averaged over the total volume of the electrode $V_{\text{electrode}}$. The volume is taken to be the volume over which local stress is averaged, which corresponds to the volume of the simulation box. Finally, a sign change is applied to yield the macroscopic pressure $\sigma_{\text{electrode}}$

$$\sigma_{\text{electrode}} = \frac{-1}{3 \cdot V_{\text{electrode}}} \left(\sum_{n=1}^{N_{\text{particles}}} \sigma_{xx} + \sigma_{yy} + \sigma_{zz} \right) \quad (8)$$

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

calendaring, discrete element method simulations, intercalation, lithium-ion electrodes, microstructure

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